Opposite numbers, and the modulus of a number

The same distance from zero, on the two sides — and the number that forgets the side.

LESSON 21–22 | 2 × 40 MIN | MATEMATIKA 7, §8 | S8 1

LESSON CLOCK

15' | 25' | 25' | 20' | 5'

Opposite numbers	15 min
The modulus	25 min
Equations with a modulus	25 min
Modulus and distance	20 min
Homework	5 min

LEARNING OBJECTIVES

- Define opposite numbers and use −(−a) = a.
- Define the modulus and compute it for any number.
- Solve simple equations of the form |x| = a.
- Use the modulus to express a distance on the coordinate line.

TEXTBOOK

National	pp. 53–58
Cambridge	Stage 8, pp. 2–10
Workbook	Exercise 1.1

01 Explanation

Opposite numbers

Two numbers are **opposite** if they lie the same distance from 0 on opposite sides. The opposite of a is written $-a$.

NUMBER	ITS OPPOSITE
7	-7
-7	7
0	0
-2.5	2.5

$$-(-a) = a \quad a + (-a) = 0$$

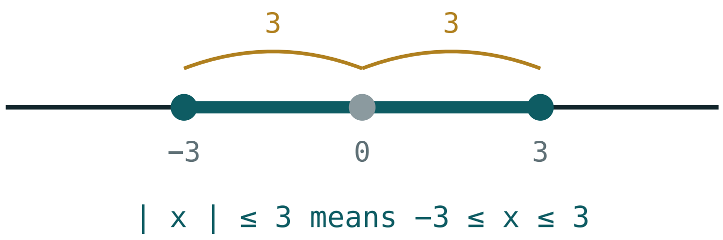
$-A$ IS NOT ALWAYS A NEGATIVE NUMBER

If $a = -7$ then $-a = 7$, which is positive. The minus sign means “the opposite of”, not “a negative number”. This is one of the hardest points of the whole chapter.

The modulus

The **modulus** $|a|$ is the distance from a to 0 on the coordinate line. Distance is never negative, so:

$$|a| = a \text{ if } a \geq 0,\ |a| = -a \text{ if } a < 0$$



The modulus is the distance to the origin — the same for a number and its opposite.

A	$\|A\|$
7	7
-7	7
0	0
-2.5	2.5

THE MODULUS FORGETS THE SIGN, AND KEEPS THE SIZE

Opposite numbers have the same modulus, which is exactly what “same distance from zero” means. So $|a| = |-a|$ for every a.

Equations with a modulus

EQUATION	SOLUTIONS	WHY
$\lvert x\rvert = 5$	$x = 5$ or $x = -5$	two points at distance 5
$\lvert x\rvert = 0$	$x = 0$	only the origin
$\lvert x\rvert = -3$	none	a distance is never negative
$\lvert x\rvert < 3$	$-3 < x < 3$	nearer than 3 to zero

$|X| = 5$ HAS TWO ANSWERS, NOT ONE

Giving only $x = 5$ loses half the marks. Asking "which points are 5 units from zero?" makes both visible immediately.

Modulus and distance

The distance between the points with coordinates a and b is

$$d = |a - b| = |b - a|$$

POINTS	WORKING	DISTANCE
$3, 10$	$\lvert 3 - 10\rvert = \lvert -7\rvert$	7
$-4, 6$	$\lvert -4 - 6\rvert = \lvert -10\rvert$	10
$-9, -2$	$\lvert -9 + 2\rvert = \lvert -7\rvert$	7

The order of subtraction does not matter, because $|a - b|$ and $|b - a|$ are opposites and therefore have the same modulus.

TWO USEFUL PROPERTIES

$|ab| = |a| \cdot |b|$ and $|a| \geq 0$ always, with equality only for $a = 0$. Both follow at once from the definition, and both will be used in every later year.

02 Worked examples

EXAMPLE 1 **Find $|-7|$, $|3.5|$, $|0|$ and $-(-4)$.**

1. $|-7| = 7$

   Seven units from zero.

2. $|3.5| = 3.5$

3. $|0| = 0$

4. $-(-4) = 4$

   The opposite of the opposite.

ANSWER

$7, 3.5, 0, 4$

EXAMPLE 2 **Solve $|x| = 6$ and $|x| = -2$.**

1. Which points are 6 from zero?

2. $x = 6$ or $x = -6$.

   Two answers.

3. A distance is never negative.

4. $|x| = -2$ has no solution.

ANSWER

$x = \pm 6$; no solution

EXAMPLE 3 **Find the distance between the points -4 and 6 on the coordinate line.**

1. $d = |-4 - 6|$
2. $= |-10|$
3. $= 10$
4. Check: 4 units to zero, then 6 more.

ANSWER

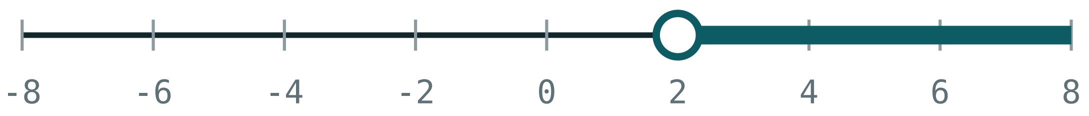
10

03 Interactive model

Ask two pupils to stand equally far from a chalk zero on opposite sides; they have opposite coordinates and the same modulus, and the definition needs no more.

Distance from zero

$$x > 2$$



inequality **x > 2** | boundary **2 (open)** | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Opposite number	Qarama-qarshi son	Противоположное число
Modulus	Modul	Модуль
Absolute value	Absolyut qiymat	Абсолютная величина
Distance from zero	Noldan masofa	Расстояние от нуля
Non-negative	Manfiy emas	Неотрицательный
Two solutions	Ikkita yechim	Два решения
Sign	Ishora	Знак
Symmetric	Simmetrik	Симметричный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The opposite of -7:

- -7
- 7
- 0
- $\frac{1}{7}$

$-(-a)$ equals:

- $-a$
- a
- 0
- $|a|$

$|-7|$ equals:

- -7
- 7
- 0
- 49

$|x| = 5$ has:

- one solution
- two solutions
- no solution
- many

$|x| = -3$ has:

- one solution
- two solutions
- no solution
- $x = 3$

The distance between a and b:

- $a - b$
- $|a - b|$
- $a + b$
- $|a| - |b|$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Opposite of 7

ANSWER -7

2. Opposite of -7

ANSWER 7

3. Opposite of 0

ANSWER 0

4. $|-7|$

ANSWER 7

5. $|3.5|$

ANSWER 3.5

6. $|0|$

ANSWER 0

7. $-(-4)$

ANSWER 4

Medium · the standard question

7 PROBLEMS

1. Solve $|x| = 6$

   ANSWER $x = \pm 6$

2. Solve $|x| = 0$

   ANSWER $x = 0$

3. Solve $|x| = -2$

   ANSWER No solution

4. Distance between -4 and 6

   ANSWER 10

5. Distance between -9 and -2

   ANSWER 7

6. $|-3| + |5|$

   ANSWER 8

7. $|-3 + 5|$

   ANSWER 2

Hard · stretch and reason

7 PROBLEMS

1. Solve $|x| < 3$ in integers

   ANSWER $-2, -1, 0, 1, 2$

2. Solve $|x| \geq 4$ in integers from -6 to 6

   ANSWER $\pm 4, \pm 5, \pm 6$

3. $|-2| \cdot |-5|$

   ANSWER 10

4. $|(-2)(-5)|$

   ANSWER 10

5. For which a is $|a| = a$?

   ANSWER $a \geq 0$

6. For which a is $|a| = -a$?

   ANSWER $a \leq 0$

7. Solve $|x - 3| = 5$

   ANSWER $x = 8$ or $x = -2$

07 Homework

Homework — 5 tasks

Every modulus question has a picture: mark the point and measure to zero.

1. Find $|-12|$, $|4.8|$, $-(-9)$ and the opposite of -15.
2. Solve $|x| = 9$ and $|x| = -1$.
3. Find the distance between -11 and 5.
4. List all integers with $|x| \leq 3$.
5. Explain why $-a$ need not be a negative number.